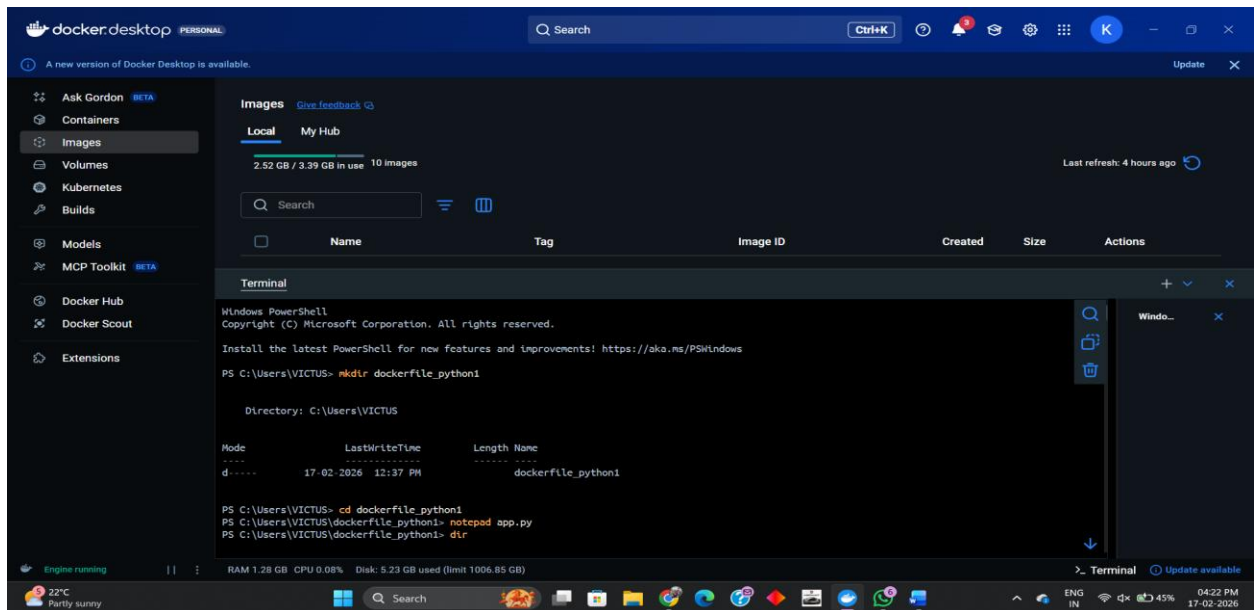


Prac-04

To create a Python Application



Step1: create a folder

Step2: create a file name app.py

from flask import Flask

app = Flask(__name__)

@app.route("/") def

home():

return "Hello from Docker!" app.run(host="0.0.0.0",

port=5000)

```
PS C:\Users\VICTUS\dockerfile_python1> notepad requirements.txt
PS C:\Users\VICTUS\dockerfile_python1> notepad dockerfile.txt
PS C:\Users\VICTUS\dockerfile_python1> dir

Directory: C:\Users\VICTUS\dockerfile_python1

Mode                LastWriteTime         Length Name
----                -
-a----             17-02-2026   12:39 PM             151 app.py
-a----             17-02-2026   12:43 PM             323 dockerfile.txt
-a----             17-02-2026   12:42 PM              5 requirements.txt

PS C:\Users\VICTUS\dockerfile_python1> ren dockerfile.txt dockerfile
PS C:\Users\VICTUS\dockerfile_python1> dir
```

Step3: create another filename requirement.txt (write flask in it)

Step4: create the dockerfile

1. Base image

FROM python:3.11-slim

2. Set working directory

WORKDIR /app

3. Copy dependency file

COPY requirements.txt

4. Install dependencies

RUN pip install --no-cache-dir -r requirements.txt

5. Copy application code

COPY . .

6. Expose port

EXPOSE 5000

7. Run the app

CMD ["python", "app.py"]

```
Directory: C:\Users\VICTUS\dockerfile_python1

Mode                LastWriteTime         Length Name
----                -
-a----           17-02-2026  12:39 PM             151 app.py
-a----           17-02-2026  12:43 PM             323 dockerfile
-a----           17-02-2026  12:42 PM              5 requirements.txt

PS C:\Users\VICTUS\dockerfile_python1> docker build -t python11
ERROR: docker: 'docker buildx build' requires 1 argument

Usage:  docker buildx build [OPTIONS] PATH | URL | -

Run 'docker buildx build --help' for more information
PS C:\Users\VICTUS\dockerfile_python1> docker build -t python11 .
[+] Building 0.2s (1/1) FINISHED                                docker:desktop-linux
```

Step5: check the directory

```
-----
ERROR: failed to build: failed to solve: failed to compute cache key: short read: expected 29778596 bytes but got 15831040: unexpected EOF
PS C:\Users\VICTUS\dockerfile_python1> docker build -t python11 .
[+] Building 256.0s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring Dockerfile: 364B
=> [internal] load metadata for docker.io/library/python:3.11-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:0b23cfb7425d065008b778022a17b1551c82f8b4866ee5a7a200084b7e2eafbf
=> => resolve docker.io/library/python:3.11-slim@sha256:0b23cfb7425d065008b778022a17b1551c82f8b4866ee5a7a200084b7e2eafbf
=> => sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 29.78MB / 29.78MB
=> => extracting sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b67c7e245da0 0.9s
=> => extracting sha256:64faa99400e1388ed9f202917ada9fac34fd46c950d40cd4102364ee9d6ab804 0.1s
=> => extracting sha256:8cbc47ff628d718fb76f7fca9897e4e8b607a4f543008cdee760705eecea1b24 0.4s
=> => extracting sha256:d85099f0969e8b2306770a12dffcd300208cc3a18b87c0ad3dc0cb51aefb9b 0.0s
=> [internal] load build context
=> => transferring context: 93B
=> [2/5] WORKDIR /app 0.1s
```

Step6: Build the image

```
=> [internal] load build context 0.0s
=> => transferring context: 93B 0.0s
=> [2/5] WORKDIR /app 0.1s
=> [3/5] COPY requirements.txt . 0.0s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt 4.6s
=> [5/5] COPY . . 0.0s
=> exporting to image 1.0s
=> => exporting layers 0.6s
=> => exporting manifest sha256:23e49e60c1df92dbb75587d7333b670e3415ab74a510e97e35fa5ae42803f3d9 0.0s
=> => exporting config sha256:e444855fdb9d2a41230b7fe3bd147f24ab9deb96dac73b5a0ccce418528ec9c9 0.0s
=> => exporting attestation manifest sha256:415c841257999998ec0576839b8afcb674c57f49f3971fa9bc5ea965675b5b10 0.0s
=> => exporting manifest list sha256:53628deadeb43fcfd17db6c1b46d4a8fa7ba772df01af8ca35ad9fefbf2d37b9 0.0s
=> => naming to docker.io/library/python11:latest 0.0s
=> => unpacking to docker.io/library/python11:latest 0.2s
PS C:\Users\VICTUS\dockerfile_python1> DOCKER IMAGES
docker: unknown command: docker IMAGES

Run 'docker --help' for more information
PS C:\Users\VICTUS\dockerfile_python1> docker images
```

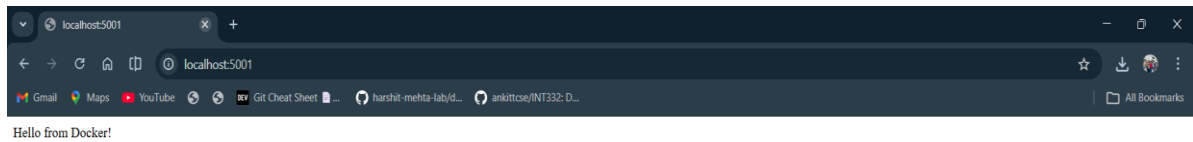
Run 'docker --help' for more information
PS C:\Users\VICTUS\dockerfile_python1> docker images

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
alpine:latest	25109184c71b	13.1MB	3.95MB	U
harpreet81/ap1:v1.1	59855d3dceb3	13MB	3.86MB	U
httpd:latest	dd178595edd6	177MB	47.6MB	U
kavya4619m/ap1:v1.1	25109184c71b	13.1MB	3.95MB	U
kavya4619m/ap2:v2.1	69eae0896851	33.6MB	10.1MB	U
mongo:latest	6f55a078e025	1.27GB	333MB	U
mysql:latest	6b18d01fb632	1.27GB	283MB	U
nginx:latest	c881927c4077	240MB	65.7MB	U
python11:latest	53628deadeb4	210MB	51.1MB	
ub1:latest	a423edda30b1	412MB	115MB	U
ubuntu:latest	cd1dba651b30	119MB	31.7MB	U

```
=> => transferring context: 2B 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 285B 0.0s
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:0b23cfb7425d065008b778022a17b1551c82f8b4866ee5a7a200084b7e2eafbf 0.0s
=> => resolve docker.io/library/python:3.11-slim@sha256:0b23cfb7425d065008b778022a17b1551c82f8b4866ee5a7a200084b7e2eafbf 0.0s
=> => sha256:632a0c2f0891418e978e637a50c920034a3a2cd9dc1b39a610f28e901c2b5ecc 0.0s
=> => naming to docker.io/library/python11:latest 0.0s
=> => unpacking to docker.io/library/python11:latest 0.0s
PS C:\Users\VICTUS\dockerfile_python1> docker rm p1
p1
PS C:\Users\VICTUS\dockerfile_python1> docker run -d --name p1 -p 5001:5000 python11
e70fb4e2d171b8e118f414cb75ad4b9c7ec6734b7fe509edd807d75c566707c1
PS C:\Users\VICTUS\dockerfile_python1> docker ps
CONTAINER ID IMAGE COMMAND STATUS PORTS NAMES
e70fb4e2d171 python11 "python app.py" 7 seconds ago Up 7 seconds 0.0.0.0:5001->5000/tcp, [::]:5001->5000/tcp p1
PS C:\Users\VICTUS\dockerfile_python1>
```

Step7: check the image build

Step8: do the port mapping



Step9: run the server on the browser